

MOLODYOZHNYAYA STATION, Antarctica

WMO: 895420

Lat: 67.666S

Lon: 45.850E

Elev: 164

StdP: 14.61

Time Zone: 3.00 (E03)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-22.2	-18.3	-30.2	1.0	-19.9	-27.2	1.2	-18.0	N/A	N/A	N/A	N/A	N/A	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	38.5	32.4	36.6	31.0	34.9	29.9	32.9	37.5	31.6	35.4	30.6	34.1	N/A	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.3	23.5	32.3	27.9	22.0	31.8	26.6	20.7	31.3	12.3	37.9	11.7	35.7	11.3	34.3	40.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-27.1	35.0	6.1	2.0	-31.5	36.4	-35.1	37.6	-38.5	38.7	-42.9	40.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	30.1	25.3	17.2	10.7	4.4	4.1	1.9	-1.1	-1.0	6.9	19.6	28.0		
	DBStd	13.28	3.61	4.85	6.77	7.92	8.61	8.62	9.52	9.76	9.44	7.69	6.31	3.93		
	HDD50	13831	617	692	1017	1180	1414	1377	1491	1584	1529	1335	912	683		
	HDD65	19306	1082	1112	1482	1630	1879	1827	1956	2049	1979	1800	1362	1148		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	17.9	1.3	1.6	1.6	1.6	1.5	1.7	1.7	1.5	1.2	1.5	1.5	1.2		
(15) Precipitation	PrecMax	27.0	3.1	3.7	2.7	2.8	3.2	4.1	3.9	2.9	2.5	4.4	3.2	2.9		
	PrecMin	11.2	0.3	0.8	0.4	0.4	0.6	0.4	0.6	0.4	0.3	0.5	0.4	0.2		
	PrecStd	4.0	0.7	0.7	0.5	0.6	0.6	0.9	0.8	0.7	0.6	0.9	0.8	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	37.8	31.4	29.1	27.5	24.3	24.9	22.5	21.1	25.7	35.4	39.9		
		MCWB	34.2	32.3	28.6	26.8	26.3	22.0	23.1	19.9	19.1	24.2	29.0	33.1		
	2%	DB	39.3	34.9	29.5	26.2	22.5	22.3	22.0	19.4	19.1	22.5	31.8	37.5		
		MCWB	32.9	29.8	27.4	24.4	21.1	20.2	20.3	17.9	17.7	20.3	27.0	31.6		
	5%	DB	37.5	33.0	27.9	24.1	20.4	20.2	19.4	16.1	16.2	20.0	29.6	35.8		
		MCWB	31.8	28.9	26.1	22.5	19.2	18.4	18.2	14.6	15.0	18.4	25.6	30.4		
	10%	DB	35.8	31.7	26.2	22.1	17.4	17.4	16.1	12.8	13.1	17.6	27.8	34.1		
		MCWB	30.5	28.1	24.4	20.3	15.7	15.6	14.4	11.4	11.7	15.9	24.4	29.3		
	0.4%	WB	35.1	32.8	29.6	27.7	25.7	22.4	23.5	20.3	19.7	24.7	29.6	33.7		
		MCDB	40.4	36.3	30.1	28.7	26.6	23.1	24.0	22.1	20.6	25.5	32.8	39.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.6	31.1	28.1	25.0	21.5	20.3	20.8	17.9	18.0	20.8	27.5	32.0		
		MCDB	38.6	33.4	29.4	26.2	22.1	21.7	22.3	19.5	19.2	22.1	30.6	36.2		
	5%	WB	32.3	29.7	26.6	22.8	19.4	18.3	18.4	14.1	15.1	18.6	25.8	30.8		
		MCDB	36.7	32.1	28.1	24.1	20.5	20.0	19.6	15.3	16.4	20.1	28.7	34.8		
	10%	WB	31.3	28.5	24.8	20.3	16.1	15.5	15.0	9.9	11.8	16.2	24.3	29.7		
		MCDB	35.2	31.2	26.2	21.9	17.5	17.3	16.3	11.1	13.0	17.6	27.1	33.5		
	5% DB	MDBR	7.3	7.8	8.3	8.5	7.6	8.2	8.1	9.4	9.4	9.7	9.2	7.4		
		MCDBR	9.2	7.8	7.1	9.1	9.1	9.1	9.0	10.2	11.5	10.1	11.5	8.9		
		MCWBR	6.4	6.1	7.1	8.3	9.0	9.0	9.0	10.3	11.5	9.4	9.1	7.1		
		MCWBR	8.4	6.7	6.5	8.1	9.1	9.4	8.7	11.4	11.7	8.9	10.1	8.0		
(40) Clear-Sky Solar Irradiance	taub	0.255	0.249	0.226	0.186	0.125	N/A	0.125	0.170	0.200	0.218	0.236	0.255			
	taud	2.234	2.282	2.303	2.164	1.901	N/A	1.879	2.107	2.216	2.218	2.228	2.205			
	Ebn at Noon	317	300	275	212	76	0	64	215	282	311	323	322			
	Edh at Noon	37	31	23	13	3	0	3	13	23	32	37	40			
(44) All-Sky Solar Radiation	RadAvg	2463	1583	820	246	33	0	10	148	644	1433	2338	2816			
	RadStd	106	102	69	19	3	0	1	8	30	50	68	85			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page